Lab Exercise 11- Deployments with Rolling Update and Recreate Strategies

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Batch-2

Understand how to use the rolling update and recreate strategies for deploying applications using Kubernetes Deployments.

Step 1: Create a Deployment with Rolling Update Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-rolling.yaml** with the following content:

```
apiVersion: apps/v1 kind:
Deployment metadata:
  name: nginx-deployment-rolling spec:
  replicas: 10
minReadySeconds: 10
selector:     matchLabels:
      app: web   strategy:
    type: RollingUpdate





    rollingUpdate:
maxUnavailable: 1
maxSurge: 5
template:
metadata:       labels:
        app: web     spec:       containers:
- name: nginx        image:
hkshitesh/kubedemo:1.0         ports:
        - containerPort: 80
```

Apply the deployment:

kubectl apply -f nginx-deployment-rolling.yaml ; watch "kubectl get rs -o wide"

```
NAME                                    DESIRED   CURRENT   READY   AGE   CONTAINERS   IMAGES                   SELECTOR
nginx-deployment-rolling-8465d4f6c7     10        10        0       66s   nginx        hkshitesh/kubedemo:1.0   app=web,po
d-template-hash=8465d4f6c7
```

Verify the deployment:

kubectl get deployments

```
PS D:\kubs> kubectl get deployments
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-rolling    2/10    10           2           104s
PS D:\kubs>
```

kubectl get pods -l app=web

```
PS D:\kubs> kubectl get pods -l app=web
NAME                                        READY   STATUS              RESTARTS   AGE
nginx-deployment-rolling-8465d4f6c7-4d5tv   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-7wm85   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-csq8x   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-fxdw6   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-hh2t5   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-jr2dm   1/1     Running             0          2m4s
nginx-deployment-rolling-8465d4f6c7-lgdsb   1/1     Running             0          2m4s
nginx-deployment-rolling-8465d4f6c7-lvdjq   0/1     ContainerCreating   0          2m4s
nginx-deployment-rolling-8465d4f6c7-s856p   1/1     Running             0          2m4s
nginx-deployment-rolling-8465d4f6c7-z2zfj   1/1     Running             0          2m4s
PS D:\kubs>
```

Update the deployment to a new image:

kubectl set image deployment/nginx-deployment-rolling nginx= hkshitesh/kubedemo:2.0

```
PS D:\kubs> kubectl set image deployment/nginx-deployment-rolling nginx=hkshitesh/kubedemo:2.0
deployment.apps/nginx-deployment-rolling image updated
PS D:\kubs>
```

Monitor the rolling update:

kubectl rollout status deploymentnginx-deployment-rolling

```
deployment.apps/nginx-deployment-rolling image updated
PS D:\kubs> kubectl rollout status deployment nginx-deployment-rolling
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 7 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 5 old replicas are pending termination...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 5 old replicas are pending termination...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 5 old replicas are pending termination...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 5 old replicas are pending termination...
```

```
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 9 of 10 updated replicas are available...
deployment "nginx-deployment-rolling" successfully rolled out
PS D:\kubs>
```

Verify the updated pods:

kubectl get pods -l app=web -o wide

```
deployment "nginx-deployment-rolling" successfully rolled out
PS D:\kubs> kubectl get pods -l app=web -o wide
NAME                                    READY   STATUS    RESTARTS   AGE    IP            NODE                    NO
MINATED NODE   READINESS GATES
nginx-deployment-rolling-565c77f48b-47z7z   1/1     Running   0          51s    10.244.0.29   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-5p56b   1/1     Running   0          115s   10.244.0.21   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-76289   1/1     Running   0          115s   10.244.0.22   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-7ds9t   1/1     Running   0          63s    10.244.0.27   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-bs24x   1/1     Running   0          115s   10.244.0.24   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-f6lgm   1/1     Running   0          51s    10.244.0.28   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-fh4dx   1/1     Running   0          115s   10.244.0.23   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-kvl5n   1/1     Running   0          115s   10.244.0.26   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-qm4dj   1/1     Running   0          50s    10.244.0.30   desktop-control-plane   <n
one>           <none>
nginx-deployment-rolling-565c77f48b-zwhjt   1/1     Running   0          115s   10.244.0.25   desktop-control-plane   <n
one>           <none>
PS D:\kubs>
```

Step 2: Create a Deployment with Recreate Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-recreate.yaml** with the following content:

```
apiVersion: apps/v1 kind:
Deployment metadata:
  name: nginx-deployment-recreate spec:
  replicas: 10   selector:
matchLabels:
      app: web
strategy:
    type: Recreate   template:





    metadata:
labels:
        app: web     spec:       containers:       -
name: nginx         image: nginx:
hkshitesh/kubedemo:1.0         ports:
        - containerPort: 80
```

Apply the deployment:

kubectl apply -f nginx-deployment-recreate.yaml ; watch "kubectl get rs -o wide"

kubectl apply -f nginx-deployment-recreate.yaml

```
while ($true) {
  kubectl get rs -o wide
  Start-Sleep -Seconds 2
  Clear-Host
}
```

```
PS D:\kubs> kubectl apply -f nginx-deployment-recreate.yaml
error: error parsing nginx-deployment-recreate.yaml: error converting YAML to JSON: yaml: line 19: mapping values are no
t allowed in this context
PS D:\kubs>
PS D:\kubs> while ($true) {
>>   kubectl get rs -o wide
>>   Start-Sleep -Seconds 2
>>   Clear-Host
>> }
NAME                                    DESIRED   CURRENT   READY   AGE     CONTAINERS   IMAGES                   SELECTOR
nginx-deployment-rolling-565c77f48b   10        10        10      3m50s   nginx        hkshitesh/kubedemo:2.0   app=web,
pod-template-hash=565c77f48b
nginx-deployment-rolling-8465d4f6c7   0         0         0       8m8s    nginx        hkshitesh/kubedemo:1.0   app=web,
pod-template-hash=8465d4f6c7
```

Verify the deployment:

kubectl get deployments

```
PS D:\kubs> kubectl get deployments
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-rolling    10/10   10           10          8m45s
PS D:\kubs>
```

kubectl get pods -l app=nginx-recreate

```
PS D:\kubs> kubectl get pods -l app=web
NAME                                    READY   STATUS    RESTARTS   AGE
nginx-deployment-rolling-565c77f48b-47z7z   1/1     Running   0          5m47s
nginx-deployment-rolling-565c77f48b-5p56b   1/1     Running   0          6m51s
nginx-deployment-rolling-565c77f48b-76289   1/1     Running   0          6m51s
nginx-deployment-rolling-565c77f48b-7ds9t   1/1     Running   0          5m59s
nginx-deployment-rolling-565c77f48b-bs24x   1/1     Running   0          6m51s
nginx-deployment-rolling-565c77f48b-f6lgm   1/1     Running   0          5m47s
nginx-deployment-rolling-565c77f48b-fh4dx   1/1     Running   0          6m51s
nginx-deployment-rolling-565c77f48b-kvl5n   1/1     Running   0          6m51s
nginx-deployment-rolling-565c77f48b-qm4dj   1/1     Running   0          5m46s
nginx-deployment-rolling-565c77f48b-zwhjt   1/1     Running   0          6m51s
PS D:\kubs>
```

Update the deployment to a new image:

```
kubectl set image deployment/nginx-deployment-recreate nginx=nginx:1.21.1
```

Monitor the update:

```
kubectl rollout status deployment nginx-deployment-recreate
```

Verify the updated pods:

```
kubectl get pods -l app=nginx-recreate -o wide
```

Step 3: Clean Up

Delete the deployments:

```
kubectl delete deployment nginx-deployment-rolling kubectl
delete deployment nginx-deployment-recreate
```

Verify that all resources are cleaned up:

kubectl get deployments

```
D:\kubs>kubectl get deployments
No resources found in default namespace.
```

kubectl get pods -l app=nginx

```
D:\kubs>kubectl get pods -l app=nginx
No resources found in default namespace.
```

kubectl get pods -l app=nginx-recreate